

ALEXANDRIA INTL, Egypt

WMO: 623180

Lat: 31.184N

Lon: 29.949E

Elev: -2

StdP: 101.35

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	8.1	1.9	4.3	13.7	3.1	4.7	13.6	12.6	13.4	11.2	13.5	1.5	190	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	33.9	22.0	32.1	23.2	31.1	23.6	25.7	30.3	25.1	29.8	24.7	29.3	4.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.1	28.7	23.8	18.6	28.4	23.1	17.8	28.1	79.2	30.2	76.8	29.8	75.0	29.2	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.5	7.7	DB	4.6	39.5	1.6	2.3	3.5	41.1	2.5	42.5	1.6	43.7	0.4	45.4	
			WB	3.6	26.9	1.4	0.8	2.6	27.5	1.7	28.0	0.9	28.5	-0.1	29.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.1	14.0	14.5	16.5	19.3	22.4	25.4	27.3	27.8	26.4	23.5	19.5	15.7	(6)
(7)		DBStd	5.27	1.73	2.11	2.38	2.72	2.43	1.77	1.15	1.10	1.52	1.94	2.23	2.04	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	423	134	108	67	17	1	0	0	0	0	0	12	84	(9)
(10)		CDD10.0	4039	124	127	202	278	384	462	536	551	493	419	285	178	(10)
(11)		CDD18.3	1420	0	2	11	45	127	212	277	292	243	161	47	3	(11)
(12)		CDH23.3	12768	3	19	96	316	757	1702	2896	3332	2416	1052	171	9	(12)
(13)	CDH26.7	3869	0	3	29	116	246	467	918	1158	712	198	21	1	(13)	
(14)	Wind	WSAvg	3.9	3.6	3.8	4.0	4.2	4.1	4.4	4.6	4.3	4.0	3.6	3.2	3.3	(14)
(15)	Precipitation	PrecAvg	161	48	29	10	2	1	0	0	1	7	28	36	(15)	
(16)		PrecMax	335	112	75	38	9	6	0	0	7	47	114	115	(16)	
(17)		PrecMin	35	0	0	0	0	0	0	0	0	0	0	0	(17)	
(18)		PrecStd	68	32	22	11	3	1	0	0	1	10	32	24	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	26.1	30.1	34.8	37.1	36.0	34.8	33.8	35.1	32.7	29.1	24.2	(19)
(20)			MCWB	14.6	16.0	18.1	19.6	20.6	21.8	23.5	24.4	23.0	21.1	19.7	17.6	(20)
(21)		2%	DB	20.2	22.4	25.8	29.8	32.0	32.1	32.0	32.1	32.2	30.0	26.5	22.1	(21)
(22)			MCWB	14.4	14.8	16.5	18.6	19.6	22.0	24.3	24.7	23.2	21.7	19.9	16.0	(22)
(23)		5%	DB	19.2	20.2	22.9	26.9	29.1	30.2	31.1	31.2	30.9	28.7	25.1	21.1	(23)
(24)			MCWB	14.0	14.3	15.9	17.6	19.8	22.5	24.1	24.6	23.4	21.6	18.9	15.7	(24)
(25)		10%	DB	18.2	18.9	20.9	24.1	26.9	29.1	30.2	30.9	29.9	27.6	24.0	20.1	(25)
(26)			MCWB	13.4	13.8	15.3	17.1	19.6	22.2	23.7	24.4	23.1	21.1	18.2	15.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	17.9	19.4	20.9	22.5	24.9	26.1	26.5	25.6	24.2	22.3	18.9	(27)
(28)			MCDWB	20.1	23.7	27.0	29.9	29.6	30.3	31.1	30.8	30.4	28.9	25.9	22.5	(28)
(29)		2%	WB	15.3	16.0	17.5	19.5	21.5	24.0	25.2	25.9	24.8	23.2	20.9	17.2	(29)
(30)			MCDWB	19.2	20.5	23.7	27.5	27.7	29.2	30.1	30.4	29.5	27.7	24.7	20.9	(30)
(31)		5%	WB	14.7	15.1	16.5	18.4	20.7	23.4	24.8	25.4	24.2	22.4	19.9	16.2	(31)
(32)			MCDWB	18.4	19.1	21.4	24.7	26.6	28.4	29.5	29.9	29.0	27.0	23.7	20.1	(32)
(33)		10%	WB	14.0	14.3	15.6	17.5	20.1	22.9	24.2	24.9	23.7	21.8	18.9	15.5	(33)
(34)			MCDWB	17.5	18.1	20.0	23.0	25.6	27.6	28.8	29.5	28.5	26.5	23.2	19.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	9.0	9.0	9.5	8.8	7.2	6.3	6.3	7.4	8.1	8.7	8.8	(35)
(36)			MCDBR	10.5	12.0	13.6	15.4	14.0	9.7	7.5	7.1	9.6	10.2	10.4	10.1	(36)
(37)			MCWBR	6.1	6.3	6.3	6.2	5.1	3.8	3.1	2.9	3.9	4.3	5.1	5.7	(37)
(38)		5% WB	MCDWB	9.8	10.4	11.4	12.9	10.5	8.0	6.9	6.6	8.2	9.2	9.5	9.2	(38)
(39)			MCWBR	6.1	5.9	5.8	5.7	4.5	3.6	3.1	3.1	3.7	4.4	5.1	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.393	0.434	0.461	0.506	0.499	0.433	0.422	0.430	0.437	0.431	0.401	0.391	(40)	
(41)		taud	2.234	2.094	2.022	1.927	1.983	2.224	2.264	2.247	2.225	2.227	2.290	2.269	(41)	
(42)		Ebn at Noon	814	812	820	798	805	856	864	854	834	807	802	794	(42)	
(43)		Edh at Noon	116	145	167	191	182	143	136	137	135	126	109	107	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.20	4.05	5.44	6.61	7.43	7.98	7.78	7.11	6.14	4.73	3.52	2.90	(44)	
(45)		RadStd	0.14	0.25	0.19	0.23	0.21	0.12	0.07	0.08	0.08	0.15	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.46	N/A	N/A	+0.62	+0.43	N/A	N/A	-68	+193	+119	(46)	(47)
(47)	Regional (4 neighbors)	+0.55	N/A	-0.65	+0.53	+0.37	N/A	N/A	-58	+199	+141	(47)	

Nomenclature: See separate page